



DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester (CE/EE/SET/ECE/ME/R&IA)

Mid Term – I Examination

Integral Calculus and Difference Equations (MAIC 102)

Max. Marks: 15 Time: 10:00 am – 10:50 am

Note:

Answer all the questions.

Calculator is not allowed into the examination hall.

- Q1. Find all the eigenvalues and eigenfunctions of Sturm-Liouville boundary value problem $y'' + \lambda y = 0$, $y(0)=y(L)=0$. (5M)
- Q2. Evaluate $\iint_R y \, dx \, dy$, where R is the region bounded by parabolas $y^2 = 4x$ and $x^2 = 4y$. (5M)
- Q3. Use Frobenius method to find series solution of $xy'' + y' - y = 0$ about $x = 0$. (5M)